

Software Development Management Lab · Lab 01

Project Kick-off & Project Charter

Software Project Management, Agile, Tool Setup and Your First Deliverable

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Why Software Project Management Matters

Good programmers can still deliver a failed project. Most failures are management failures, not bad code.

- Unclear or constantly changing requirements
- Unrealistic estimates and deadlines
- Poor communication between stakeholders
- Scope creep with no defined boundaries

Management keeps a project on time, on budget and at the right quality.

Lessons from Real Project Failures

Denver Airport Baggage System

Delayed the airport's opening by 16 months. Complexity was underestimated and integration testing was weak.

Healthcare.gov Launch (2013)

Crashed on launch day. Too little load testing, unclear contractor ownership and a compressed timeline.

FBI Virtual Case File

About \$170M spent before it was scrapped. Requirements kept shifting and there was no firm scope.

Discussion: If you were the project manager, what would you have done first?

SDLC and Process Models: A Quick Recap

SDLC Phases

1. Planning
2. Requirement Analysis
3. Design
4. Development
5. Testing
6. Deployment
7. Maintenance

Popular Models

- Waterfall
- Spiral
- Iterative
- Agile

This lab focuses on Agile practices.

Agile: The Four Values

Individuals and interactions

over

processes and tools

Working software

over

comprehensive documentation

Customer collaboration

over

contract negotiation

Responding to change

over

following a plan

Small releases (sprints), customer feedback and continuous improvement.

We will practise: Scrum · XP · Lean · Kanban

Our Toolchain for the Semester

Version Control: GitHub

Shared team repository, branches and pull requests

Project Board: Trello / Jira

Team board; later our backlog and sprint board

Scheduling: GanttProject

Free tool for Gantt charts and the critical path (or ProjectLibre)

Docs & Chat

Google Docs / Notion for documents; Teams / Slack for talking

Set these up today. Every later lab depends on them.

Task A: Set Up Your Tools

1. One member creates a GitHub repo named sdm-teamname
2. Add all members and the instructor as collaborators
3. Create a Trello or Jira board with 4 columns: Backlog | To Do | In Progress | Done
4. Create a shared Google Doc or Notion page for documents
5. Install GanttProject or ProjectLibre for later labs

Check: everyone can open the repo, the board and the doc.

Choose Your Semester Project

Online Food Ordering

Customers order and pay online; staff manage menus, orders and branches.

Library Management

Members search, borrow and return books; librarians manage stock and fines.

Hospital Appointments

Patients book appointments; doctors manage schedules and patient history.

E-Learning Platform

Students enroll and submit work; instructors upload content and grade.

Or propose your own idea, with the instructor's approval.

What Is a Project Charter?

A short document that formally starts a project and gives the whole team one shared definition of it.

WHY?

What problem are we solving? What is the business need?

WHAT?

What is in scope and out of scope? What does "done" look like?

WHO?

Who are the stakeholders, and what role does each play?

The backlog, schedule and risk register in later labs all trace back to this charter.

Charter Sections 1 to 5

1. Project Title	A short, descriptive name
2. Description	What the system does and who it serves (2–3 sentences)
3. Business Need	The problem or pain point being solved
4. SMART Objectives	Specific and measurable, e.g. "cut order time from 15 to 5 minutes within 3 months"
5. Scope: In / Out	Features delivered vs. features explicitly excluded (stops scope creep)

Charter Sections 6 to 10

6. Stakeholders	Sponsor, end users, dev team, third parties and their roles
7. Assumptions	Things believed true, e.g. "the client will provide the payment API"
8. Constraints	Fixed limits: budget, deadline, team size, required technology
9. Milestones	Requirements → Design → Development → Testing → Deployment (rough durations)
10. Success Criteria	How success will be measured, tied back to the objectives

Worked Example: Online Food Ordering

Problem Statement

A local restaurant chain wants a web and mobile app so customers can browse menus, order and pay online. Staff need a dashboard to manage orders, update menus and track daily sales. It must support many branches, live order status and a payment gateway.

Stakeholders

- Chain owner (sponsor)
- Branch managers (users)
- Customers (users)
- Your group (dev team)
- Payment gateway (third party)

Other projects: write your own problem statement and stakeholders in this format first.

Common Mistakes to Avoid

WEAK: "Make the system fast and user-friendly."

BETTER: "Average page load time under 2 seconds."

WEAK: Leaving "out of scope" blank.

BETTER: List 2–3 features excluded from version 1.

WEAK: Mixing up assumptions and constraints.

BETTER: Assumption = believed true.
Constraint = fixed limit.

Task B: Draft Your Project Charter

1. Form groups of 3–4 and assign roles: Project Manager, Developers, Tester / Documenter
2. Confirm your project topic
3. Write the problem statement and stakeholder list
4. Fill in all 10 charter sections in your shared doc
5. Save it as /docs/charter.md in your team repo
6. Prepare a 3-minute presentation of your charter

Keep objectives SMART and scope statements specific.

Deliverable and Evaluation

Due Before Lab 2

- 1–2 page Project Charter, all 10 sections
- All members listed with their roles
- Submitted as /docs/charter.md or a shared Google Doc
- Repo and board shared with the instructor

Semester Project Is Judged On

- Project idea: creativity and feasibility
- Teamwork
- Documentation
- Implementation
- Final presentation

Coming Up: Lab 2

Product Backlog and User Stories

- Turn your in-scope features into user stories
- Format: As a [user], I want [goal] so that [benefit]
- INVEST criteria and acceptance criteria
- Prioritize with MoSCoW (Must, Should, Could, Won't)

Bring your charter: its scope is the starting point.

Key Takeaways

- **Software development is not just coding**
- **Good management is the key to project success**
- **The charter defines why, what and who**
- **This lab simulates real industry teamwork**

Questions?